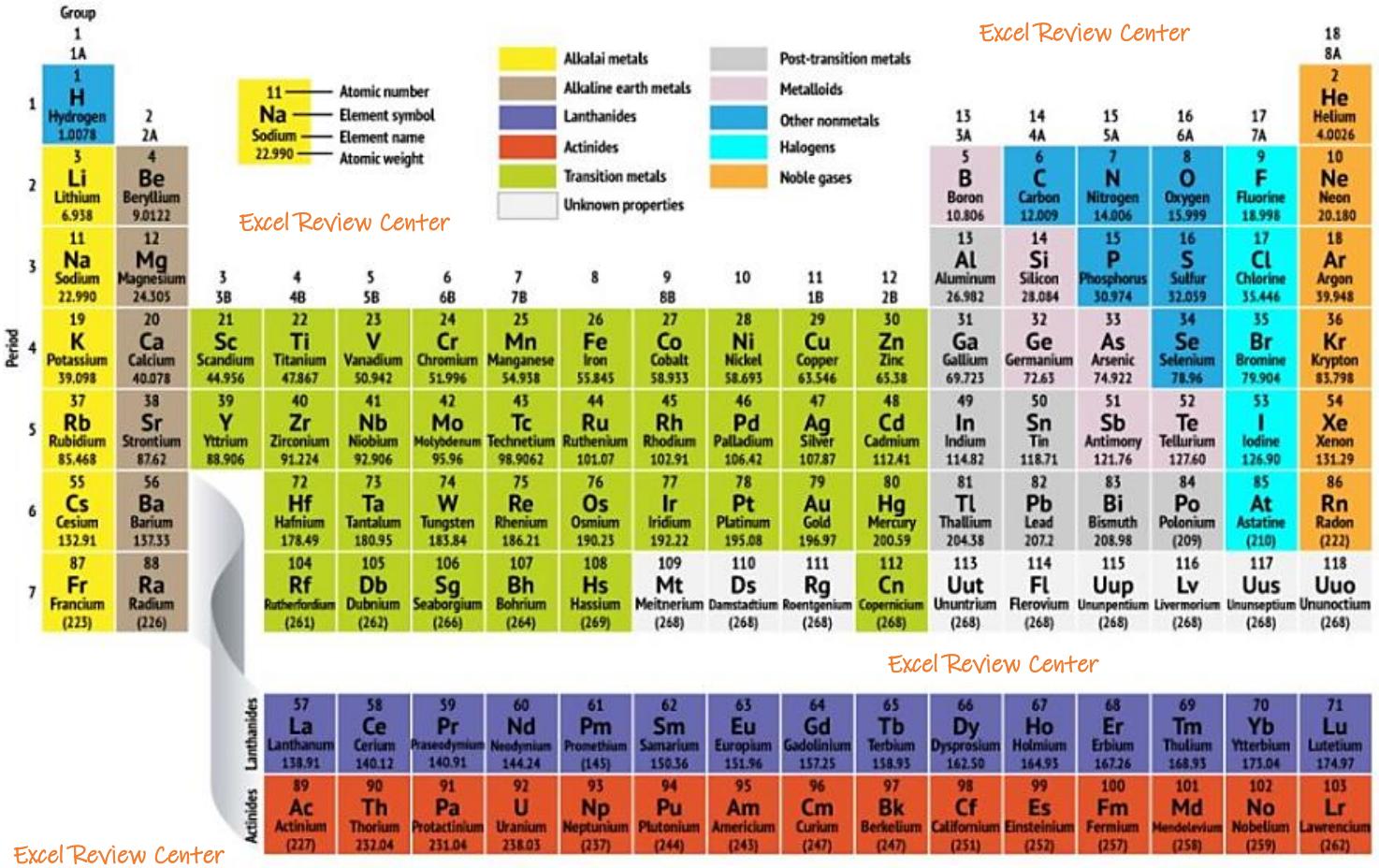
Periodic Table of the Elements



The **Periodic Table** is a chart showing the different elements arranged by their atomic numbers. The horizontal rows in the table are called periods while the vertical columns are called groups or families. Naturally, because of this arrangement, elements in the periodic table are grouped according to their properties. Nonmetals are found in the right hand part of the table. Most of the elements though in the table are classified as metals, which are found in the left part and central part of the table. Between them are elements known as metalloids. These are elements with properties that are intermediate between metals and non-metals. The periodic table can also be divided into representative elements, transition elements, noble gases, lanthanides and actinides. Excel Review Center

Representative Elements

These are elements from Group 1A to Group 7A and are found in the leftmost and rightmost part of the table. Another term used is Main Group Elements. The group number also gives out the number of the atom's valence electrons, which are the outermost electrons of an atom and are the electrons that are involved in chemical bonding. For convenience in making a reference to a representative element, groups or names are conventionally being named as follows: →

Group Number	Group Name
1A	Alkali Metals
2A	Alkaline Earth Metals
3A	Boron Group
4A	Carbon Group
5A	Nitrogen Group
6A	Chalcogens
7A	Halogens
8A	Noble Gases/ Rare Gases

Transition Elements

These are elements found at the center of the table. Another term used in Transition Metals. These are elements that can have more than 8 electrons in their shells. For instance, Gold (Au) has 18 electrons in its third shell. These elements can also use their two outermost shells when it bonds with other elements. Excel Review Center

Noble Gases

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These are elements (except for Helium) found at the rightmost column of the table. Since all of these elements have full outermost shells, they rarely combine with other elements. Hence, they are also called inert gases.

Lanthanides

these are 15 elements that are found from one of two rows that are usually separated from the periodic table. It starts with Lanthanum (La) with an atomic number of 57 up to Lutetium (Lu) with an atomic number of 71. Sometimes, these elements are called rare-earth elements or inner transition metals. These elements are found naturally in earth and of the 15 elements, only one is radioactive. Excel Review Center

Actinides

these are elements that belong to the row below the lanthanides in the periodic table. Most of these elements are radioactive and are not naturally-occuring. In other words, they are just produced in the laboratory. It starts with Actinium (Ac) with an atomic number of 89 up to Lawrencium (Lr) with an atomic number of 103.

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